

10.5.9

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Question

A straight line L is perpendicular to the line $5x - y + 1 = 0$. The area of the triangle formed by line L and the coordinate axes is 5.

Find the equation of the line L .

Step 1: Define the Lines

Let the given line be L_1 and the required perpendicular line be L .

The equation of the given line L_1 is $5x - y = -1$.

In matrix form $\mathbf{n}^\top \mathbf{x} = c$, its parameters are:

$$\mathbf{n}_1 = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad c_1 = -1 \quad (0.1)$$

Let the required line L have the equation $\mathbf{n}_2^\top \mathbf{x} = c_2$.

Step 2: Find the Normal Vector of Line L

Since the lines are perpendicular, the dot product of their normal vectors must be zero:

$$\mathbf{n}_1^\top \mathbf{n}_2 = 0$$

Let's assume the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ m \end{pmatrix}$. We solve for m :

$$\mathbf{n}_1^\top \mathbf{n}_2 = (5 \quad -1) \begin{pmatrix} 1 \\ m \end{pmatrix} = 5 - m = 0 \quad (0.2)$$

$$\implies m = 5 \quad (0.3)$$

Therefore, the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$, and its equation is:

$$x + 5y = c_2$$

Step 3: Finding the Intercepts

The line L forms a triangle with the coordinate axes.

The y-axis equation is $x = 0$, or $\mathbf{e}_1^\top \mathbf{x} = 0$, where $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

The x-axis equation is $y = 0$, or $\mathbf{e}_2^\top \mathbf{x} = 0$, where $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

We find the intercepts by solving for the intersection of line L with each axis.

Step 4: Calculating the x-intercept (**a**)

The x-intercept, **a**, is the intersection of line L ($x + 5y = c_2$) and the x-axis ($y = 0$).

We solve the stacked system:

$$\begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} c_2 \\ 0 \end{pmatrix} \quad (0.4)$$

The intercept vector **a** is found by inverting the matrix:

$$\mathbf{a} = \begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} c_2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c_2 \\ 0 \end{pmatrix} = \begin{pmatrix} c_2 \\ 0 \end{pmatrix} \quad (0.5)$$

Step 5: Calculating the y-intercept (**b**)

The y-intercept, **b**, is the intersection of line L ($x + 5y = c_2$) and the y-axis ($x = 0$).

We solve the stacked system:

$$\begin{pmatrix} 1 & 5 \\ 1 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} c_2 \\ 0 \end{pmatrix} \quad (0.6)$$

The intercept vector **b** is:

$$\mathbf{b} = \begin{pmatrix} 1 & 5 \\ 1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} c_2 \\ 0 \end{pmatrix} = \frac{1}{-5} \begin{pmatrix} 0 & -5 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} c_2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ c_2/5 \end{pmatrix} \quad (0.7)$$

Step 6: Using the Area to Find c_2

The area of the triangle formed by the origin, $\mathbf{a}$, and $\mathbf{b}$ is given as 5. Since the intercepts lie on the axes, the angle θ between them is 90° .

$$\text{Area} = \frac{1}{2} \|\mathbf{a}\| \|\mathbf{b}\| \sin(90^\circ) = 5 \quad (0.8)$$

$$\frac{1}{2} \left(\sqrt{c_2^2} \right) \left(\sqrt{(c_2/5)^2} \right) (1) = 5 \quad (0.9)$$

$$\frac{1}{2} |c_2| \left| \frac{c_2}{5} \right| = 5 \quad (0.10)$$

$$\frac{|c_2^2|}{10} = 5 \quad (0.11)$$

$$c_2^2 = 50 \implies c_2 = \pm 5\sqrt{2} \quad (0.12)$$

Solution: The Equations of Line L

There are two possible lines that satisfy the given conditions.

Substituting the values of c_2 into the equation $\mathbf{n}_2^\top \mathbf{x} = c_2$, the possible equations for line L are:

$$x + 5y = 5\sqrt{2}$$

$$x + 5y = -5\sqrt{2}$$

In matrix form:

$$(1 \ 5) \begin{pmatrix} x \\ y \end{pmatrix} = 5\sqrt{2} \quad (0.13)$$

$$(1 \ 5) \begin{pmatrix} x \\ y \end{pmatrix} = -5\sqrt{2} \quad (0.14)$$

Plot

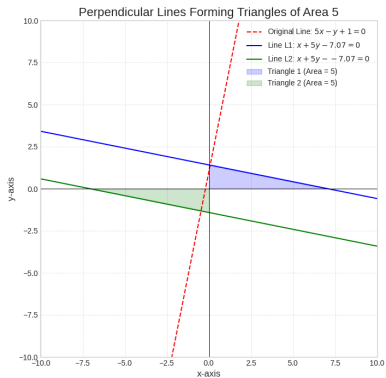


Fig :